Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: An International Prospective Open-label, Multi-center, Randomized, Non-comparative Phase II Study of Lutetium [177Lu] Vipivotide Tetraxetan (AAA617) Alone and Lutetium [177Lu] Vipivotide Tetraxetan (AAA617) in Combination With Androgen Receptor Pathway Inhibitors in Patients With PSMA PET Scan Positive Castration-Resistant Prostate Cancer **Short Title/ Acronym:** A Phase II Study of AAA617 Alone and AAA617 in Combination With ARPI in Patients With PSMA PET Scan Positive CRPC (PSMACare) **Protocol Number:** CAAA617B12203 **NCT Number:** NCT05849298 **Posting ID:** 49405554200369064 **Trial Status:** Recruiting **Sponsor/Company Name:** Novartis Pharmaceuticals / Novartis Pharma AG **Investigational Product:** Lutetium [177Lu] vipivotide tetraxetan (AAA617 / Pluvicto) and ARPIs (apalutamide, enzalutamide, or darolutamide) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy based on PSA response of Lutetium [177Lu] vipivotide tetraxetan (AAA617) alone and in combination with an Androgen Receptor Pathway Inhibitor (ARPI) in participants with PSMA-positive, castration-resistant prostate cancer (CRPC) and no evidence of metastasis in conventional imaging. **Secondary Objectives:** To evaluate Metastatic Free Survival (MFS), Radiographic Progression Free Survival (rPFS), and Overall Survival (OS).

2.2 Background & Rationale To address the therapeutic gap in non-metastatic (by conventional imaging) but PSMA PET-positive Castration-Resistant Prostate Cancer (Prostatic Neoplasm). While ARPIs and Androgen Deprivation Therapy (ADT) are the standard of care, resistance often develops. Targeted radioligand therapy with 177Lu-PSMA-617 has proven highly efficacious in late-stage metastatic CRPC. This study investigates earlier intervention—utilizing 177Lu-PSMA-617 alone or combined with ARPIs—to extend survival and improve overall clinical outcomes in high-risk patients.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (AAA617 alone vs. AAA617 + ARPI) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target N: Approximately 80 participants **Number of Sites:** International, Multi-center **Trial Status:** Recruiting **Estimated Study Start:** January 2024 **Estimated Primary Completion:** January 2029

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants must be adults ≥ 18 years of age with signed informed consent prior to participation to study. 2. Histologically or cytologically confirmed prostate cancer. 3. Ongoing androgen deprivation therapy (ADT) with a GnRH agonist/antagonist or prior bilateral orchiectomy at the time of randomization. Intermittent administration of ADT is accepted before randomization if criterion for serum testosterone is met. 4. Castrate level of serum testosterone (< 1.7 nmol/l [50 ng/dl]) on GnRH agonist or antagonist therapy (continuous/intermittent) or after bilateral orchiectomy prior to randomization. 5. Evidence of PSMA-positive disease (N1 or M1) as seen on a AAA517 or piflufolastat F 18 PET/CT scan at baseline as determined by Blinded Independent Central Review (BICR) based on the methodology proposed in the Prostate Cancer Molecular Imaging Standardized Evaluation (PROMISE). Participants with M1 disease only on PSMA PET scan are allowed to participate. 6. Participants must have a negative conventional imaging for M1 disease. 7. Participants must have adequate organ functions: bone marrow reserve, hepatic & renal.

4.2 Exclusion Criteria 1. Prior or present evidence of metastatic disease as assessed by CT/MRI locally for soft tissue disease and whole-body radionuclide bone scan for bone disease. Exception: Participants with pelvic disease may be eligible (e.g., participants with enlarged lymph nodes below the bifurcation of common iliac arteries (N1)). 2. Unmanageable concurrent bladder outflow obstruction or urinary incontinence. (Note: participants with obstruction or incontinence which is manageable with standard of care are allowed). 3. Active clinically significant cardiac disease; history of seizure or condition that may pre-dispose to seizure which may require treatment with surgery or radiation therapy. 4. Prior therapy with: second generation anti-androgens or CYP17 inhibitors < 3 months before randomization; ketoconazole (short duration <28 days permitted); radiopharmaceutical agents (e.g., Strontium-89) if wash-out period of at least 3 months is not completed; PSMA-targeted radioligand therapy; immunotherapy; chemotherapy (except if administered in the adjuvant/neoadjuvant setting completed > 2 years before randomization); any other investigational agents for CRPC; use of estrogens, 5-α reductase inhibitors, other steroidogenesis inhibitors or first-generation anti-androgens within 28 days before randomization; external beam radiation therapy [EBRT] and brachytherapy within 28 days before randomization. 5. Other concurrent cytotoxicity chemotherapy, immunotherapy, radioligand therapy, poly adenosine diphosphateribose polymerase (PARP) inhibitor, biological therapy or investigational therapy.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 7.4 GBq ($\pm$ 10%) of AAA617 administered once every 6 weeks for 6 cycles. ARPIs (apalutamide, enzalutamide, or darolutamide) are administered per standard continuous daily dosing. **Route of Administration:** AAA617 (Intravenous infusion); ARPIs (Oral) **Duration of Treatment:** 36 weeks (6 cycles of 6 weeks) for AAA617; ARPIs continuously as prescribed.

5.2 Concomitant & Prohibited Therapy Permitted: Standard ADT (GnRH agonist/antagonist or prior bilateral orchiectomy), best supportive care. **Prohibited:** Concurrent cytotoxic chemotherapy, immunotherapy, PARP inhibitors, biological therapy, or other investigational therapies.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, PSMA PET/CT scan, Conventional Imaging, Labs, Organ Function verification
Baseline	Day 0	Randomization, First Dose of AAA617 $\pm$ ARPI initiation
Interim	Every 6 Weeks	AAA617 Infusion (Cycles 1-6), Safety Labs, Efficacy Assessment, PSA measurement
End of Study	Follow-up	Long-term tracking of PSA, MFS, rPFS, OS, and adverse events

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint PSA response: PSA response is defined as the time of PSA nadir value of =< 0.2 ng/mL is confirmed by a second (the next) PSA measurement at least 4 weeks later.

7.2 Secondary Endpoints 1. Metastatic Free Survival (MFS) 2. Radiographic Progression Free Survival (rPFS) 3. Overall Survival (OS)

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits, focusing heavily on radioligand-associated toxicities (e.g., myelosuppression, dry mouth/eyes, fatigue) and ARPI-related effects. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):**

Independent safety monitoring mechanisms structured by the sponsor.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N \approx 80$, appropriately powered to detect clinically meaningful differences in PSA response and survival event rates between the combination and monotherapy arms. **Handling of Missing Data:** Censoring applied at the date of the last adequate tumor assessment for time-to-event endpoints.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of clinical data, with intense scrutiny on adherence to radiation dosing protocols. **Audit Trail:** Validated eCRF locking, tracking of Blinded Independent Central Review (BICR) imaging reads based on PROMISE methodology, and standard data clarification mechanisms.

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11. Study Identifiers & Governance

Primary ID: CAAA617B12203 **Registry Number:** NCT05849298 **Posting ID:** 49405554200369064 **Trial Status:** Recruiting **Sponsor/Lead Organization:** Novartis **Short Title:** PSMACare / A Phase II Study of AAA617 Alone and AAA617 in Combination With ARPI in Patients With PSMA PET Scan Positive CRPC **Official Regulatory Title:** An International Prospective Open-label, Multi-center, Randomized, Non-comparative Phase II Study of Lutetium [177Lu] Vipivotide Tetraxetan (AAA617) Alone and Lutetium [177Lu] Vipivotide Tetraxetan (AAA617) in Combination With Androgen Receptor Pathway Inhibitors in Patients With PSMA PET Scan Positive Castration-Resistant Prostate Cancer

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12. Clinical Framework (CRPC Specific)

Disease Area / Conditions: Prostatic Neoplasm / Castration-Resistant Prostate Cancer (CRPC) **Diagnosis Threshold:** PSMA-positive disease by PET/CT with castrate serum testosterone (< 1.7 nmol/L) **Medical Methodology:** Targeted Radioligand Therapy (RLT) combined with Androgen Receptor Pathway Inhibition **Optimization Target:** Achievement of PSA response and prolonged metastatic-free survival

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13. Study Procedures & Methodology

Management Pathway: PSMA-targeted radioligand infusion with continuous androgen suppression **Education Protocol (HCP):** Radiation safety handling, dosimetry protocols, and administration guidelines for 177Lu **Education Protocol (Patient):** Radiation precautions post-infusion and tracking of specific side effects **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of imaging scans utilizing the PROMISE methodology

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14. Observation & Follow-Up Schedule

Total Duration: Approximately 5 years for survival tracking (MFS/OS) **Onsite Visit Cadence:** Every 6 weeks during the 36-week active treatment period **Remote Monitoring Frequency:** Periodic telemedicine and localized lab follow-ups post-treatment **Checklist Review Interval:** Every 12 weeks during long-term survival follow-up

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]